Saraswat Academy
Class-X
Chapter-1 Real Numbers
Class Test

Multiple Choice Questions

Q1: The HCF of 306 and 657 is:

A. 3 B. 9 C. 18 D. 27

Q2: If the prime factorization of a number is $2^3 \times 5^2$ Then the number is:

A. 100 B. 200 C. 150 D. 250

Q3: The product of HCF and LCM of two numbers is equal to:

A. Sum of numbers B. Difference of numbers
C. Product of numbers D. Square of numbers

Q4: If the HCF of 65 and 117 is expressed in the form $65x + 117y$, then the value of HCF is:

A. 1 B. 13 C. 5 D. 3

Q5: If two numbers are coprime, then their HCF and LCM relationship is:

A. HCF = 1 B. LCM = product of numbers
C. Both A and B D. Neither

Q6: The number of prime factors in the prime factorization of 864 is:

A. 4 B. 5 C. 6 D. 7

Q7: **Assertion (A):** Every composite number has a unique prime factorization.
Reason (R): Prime numbers can be written as a product of two smaller natural numbers.

Options:
A. Both A and R are true and R explains A
B. Both A and R are true but R does not explain A
C. A is true but R is false
D. A is false but R is true

Q8: **Assertion (A):** If aand bare prime numbers, their HCF is always 1.
Reason (R): Prime numbers have only two factors: 1 and itself.

Options:
A. Both true and R explains A
B. Both true but R not explanation
C. A true R false
D. A false R true

Short and Long answer type questions:

Q11: Prove that $\sqrt{5}$ is an irrational number.

Q12: Prove that $\frac{1}{2\sqrt{7}}$ is an irrational number.

Q13: Prove that $\sqrt{5} + \sqrt{7}$ is an irrational number.

Q14: Karan has 180 blue marbles and 150 red marbles. He wants to pack them into packets containing equal number of marbles of the same color. What is the maximum number of marbles that each packet can hold?

Q15: Katya has 49 paintings and 35 medals. She wants to display them in groups throughout her house, each with the same combination of paintings and medals, with none left over. What is the greatest number of groups Katya can display?

Q9: CASE STUDY-1:

A factory produces 840 biscuits and 1260 chocolates.

The manager wants to pack them into identical boxes so that:

- Each box contains equal number of biscuits
- Each box contains equal number of chocolates
- No item remains unused.

Q1: Find the maximum number of boxes that can be made.

Q2: Find the number of biscuits in each box.

Q3: Find the of chocolates in each box.

Q10: CASE STUDY-2

Two machines are used in road construction.

- Machine A finishes a cycle every 24 minutes
- Machine B finishes a cycle every 36 minutes

Both machines start working at the same time.

Q1: After how many minutes will both machines finish a cycle together again?

Q2: How many cycles will machine A complete in that time?

Q3: How many cycles will machine B complete?